AP PRECALCULUS · TOPIC 1.10

Rational Functions and Holes

Section 1 — Multiple Choice
Circle the letter of the correct answer **SOLVED**

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 Let $r(x) = \dfrac{x^2 + 2x - 8}{x^2 - 3x + 2}$. What are the coordinates of the hole in the graph of r?

A. $(2, 6)$ **B.** $(2, 0)$

C. $(-2, 6)$ **D.** $\left(2, \dfrac{1}{6}\right)$

A. $r(x) = \dfrac{(x+4)(x-2)}{(x-1)(x-2)}$: the factor $x-2$ is in both parts, so the hole is at $x = 2$, and the reduced value is $\dfrac{2+4}{2-1} = 6$. B: used the original numerator, which is 0 at $x = 2$. C: sign error, reading $x - 2$ as $x = -2$. D: inverted the reduced fraction.

2 Which statement about the graph of $r(x) = \dfrac{x^2 - 9}{x^2 - 6x + 9}$ is true?

A. A hole at $x = 3$ and no vertical asymptote **B.** A vertical asymptote at $x = 3$ and no hole

C. A hole at $x = 3$ and a vertical asymptote at $x = -3$ **D.** A hole at $x = -3$ and a vertical asymptote at $x = 3$

B. $r(x) = \dfrac{(x-3)(x+3)}{(x-3)^2}$. At $x = 3$ the numerator has multiplicity 1 and the denominator multiplicity 2; since $1 < 2$ there is no hole, and $r(x) = \dfrac{x+3}{x-3}$ is undefined with unbounded outputs at $x = 3$. A: cancelled one factor and stopped, without comparing multiplicities. C: treated $x + 3$ as an asymptote factor, although it is in the numerator only. D: sign error and swapped the two roles.

3 What is the value of $\lim\limits_{x \to -3} \dfrac{x^2 + 2x - 3}{x + 3}$?

A. 0 **B.** 2

C. -4 **D.** The limit does not exist

C. $\dfrac{(x+3)(x-1)}{x+3} = x - 1$ for $x \neq -3$, so the limit is $-3 - 1 = -4$. A: substituted $x = -3$ into the numerator, which is 0. B: evaluated $x - 1$ at $x = 3$. D: concluded the limit fails because the denominator is 0 at $x = -3$.

4 For a rational function r, the outputs for inputs near 2 are shown. Which statement is supported by the table?

x	1.9	1.99	1.999	2.001	2.01	2.1
$r(x)$	4.9	4.99	4.999	5.001	5.01	5.1

A. The graph of r has a vertical asymptote at $x = 2$ **B.** The graph of r has a hole at $(5, 2)$

C. The graph of r has a zero at $x = 2$ **D.** The graph of r has a hole at $(2, 5)$

D. The outputs approach 5 from both sides as $x \to 2$, so $\lim\limits_{x \to 2} r(x) = 5$ and the hole is at $(2, 5)$ (here $r(x) = x + 3$ with $x \neq 2$). A: confused a hole with an asymptote (outputs would be unbounded). B: swapped the coordinates. C: read the input 2 as a zero.

5 Which function has a hole at $x = 1$ and a vertical asymptote at $x = -2$?

A. $r(x) = \dfrac{(x-1)(x+3)}{(x-1)(x+2)}$ **B.** $r(x) = \dfrac{(x+1)(x+3)}{(x+1)(x-2)}$

C. $r(x) = \dfrac{(x+2)(x+3)}{(x-1)(x+2)}$ **D.** $r(x) = \dfrac{x+3}{(x-1)(x+2)}$

A. The hole needs the factor $x - 1$ in both numerator and denominator; the asymptote needs $x + 2$ in the denominator only. B: sign errors (hole at -1, asymptote at 2). C: swapped the roles (hole at -2, asymptote at 1). D: no common factor, so both denominator zeros are asymptotes.

6 Let $r(x) = \dfrac{(x-2)^3(x+1)}{(x-2)^2(x-5)}$. What are the coordinates of the hole in the graph of r?

A. $(2, 3)$ **B.** $(2, 0)$

C. $(-1, 0)$ **D.** $(5, 0)$

B. At $x = 2$ the multiplicities are $3 \geq 2$, so there is a hole. The reduced function is $\dfrac{(x-2)(x+1)}{x-5}$, which is 0 at $x = 2$. A: evaluated only the factor $x+1$ at $x = 2$, ignoring the rest of the reduced function. C: gave a zero of the numerator instead of the hole. D: gave the asymptote's input with the y-value 0.

7 How many holes and how many vertical asymptotes does the graph of $r(x) = \dfrac{(x+1)(x-2)(x-4)}{(x-2)(x-4)(x+5)}$ have?

A. 0 holes and 3 vertical asymptotes **B.** 1 hole and 2 vertical asymptotes

C. 2 holes and 1 vertical asymptote **D.** 3 holes and 0 vertical asymptotes

C. The factors $x-2$ and $x-4$ are in both parts, giving holes at $x = 2$ and $x = 4$; the factor $x+5$ stays in the denominator, giving the asymptote $x = -5$. A: treated every zero of the denominator as an asymptote. B: cancelled only one common factor. D: counted every factor of the numerator as a hole.

8 What are the coordinates of the hole in the graph of $r(x) = \dfrac{2x^2 - 8}{x^2 - x - 2}$?

A. $(2, 2)$ **B.** $\left(2, \dfrac{4}{3}\right)$

C. $\left(2, \dfrac{3}{8}\right)$ **D.** $\left(2, \dfrac{8}{3}\right)$

D. $r(x) = \dfrac{2(x-2)(x+2)}{(x-2)(x+1)}$, so the hole is at $x = 2$ with reduced value $\dfrac{2(2+2)}{2+1} = \dfrac{8}{3}$. A: used the ratio of leading coefficients, the horizontal asymptote. B: dropped the constant factor 2. C: inverted the reduced fraction.

AP PRECALCULUS · TOPIC 1.10 · SECTION 2

Rational Functions and Holes

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

A sprinter's distance from the start, in metres, is $s(t) = t^2 + 3t$ for t seconds after the start. The average speed between 1 second and t seconds is $v(t) = \dfrac{s(t) - s(1)}{t - 1} = \dfrac{t^2 + 3t - 4}{t - 1}$, for $t > 0$ and $t \neq 1$.

(a) Factor the numerator, write $v(t)$ in simplified form and give the coordinates of the hole in the graph of v. DOK3 2 pt

$t^2 + 3t - 4 = (t + 4)(t - 1)$, so $v(t) = t + 4$ for $t \neq 1$. Hole at $(1, 5)$.

(b) Use limit notation to describe the behaviour of $v(t)$ near $t = 1$, and interpret the result in context. DOK3 2 pt

$\lim_{t \to 1} v(t) = 5$. The average speed over shorter and shorter intervals around 1 second approaches 5 metres per second.

(c) Find the time at which the average speed is 7.5 metres per second. Explain why no time gives an average speed of exactly 5 metres per second. DOK3 2 pt

$t + 4 = 7.5$, so $t = 3.5$ seconds. $v(t) = 5$ only for $t = 1$, which is the hole: $v(1)$ is undefined, so 5 is approached but never reached.

Question 10 No calculator · 6 points

Let $r(x) = \dfrac{(x - 3)^2(x + 1)(x - 2)}{(x - 3)(x - 2)^2(x + 4)}$.

(a) For each of the inputs $x = 3$ and $x = 2$, compare the multiplicity of the zero in the numerator with its multiplicity in the denominator, and state whether the graph has a hole or not. DOK3 2 pt

$x = 3$: numerator multiplicity 2, denominator multiplicity 1; $2 \geq 1$, so the graph has a hole at $x = 3$. $x = 2$: numerator multiplicity 1, denominator multiplicity 2; $1 < 2$, so there is no hole (a vertical asymptote).

(b) Find the coordinates of the hole. DOK3 2 pt

Reduced function $\dfrac{(x - 3)(x + 1)}{(x - 2)(x + 4)}$. At $x = 3$ the value is 0, so $\lim_{x \to 3} r(x) = 0$ and the hole is $(3, 0)$.

(c) State the equations of all vertical asymptotes and the x-intercepts of the graph of r. DOK3 2 pt

Vertical asymptotes: $x = 2$ and $x = -4$. The only x-intercept is $(-1, 0)$; $x = 3$ is a hole, not an intercept.

Answer key (Section 1): 1A 2B 3C 4D 5A 6B 7C 8D